

1 Search Models

1.1 Exercise 7.1

The Bellman equation for a McCall-model with a chance of unemployment is as follows

$$\begin{aligned} \bar{V} &= c + \beta \left(\phi \bar{V} + (1 - \phi) \mathbb{E} [v(w')] \right) \\ v(w) &= \max_{a \in \{\text{Accept}, \text{Reject}\}} \left\{ \frac{w}{1 - \beta}, \beta(\phi(c + \bar{V}) + (1 - \phi) \mathbb{E} [v(w')]) \right\} \end{aligned} \quad (1.1)$$

where $\bar{V}$ denotes the continuation value of being unemployed. $v(w)$ denotes the value function conditional on receiving an offer. $\mathbb{E} [v(w')] = \int_0^B v(w') dF(w')$ denotes the expected value conditional on receiving an offer.

1.2 Exercise 7.2

A McCall Model with two offers per period. Since $F(x) = \Pr(z \leq x)$, the cumulative distribution function of the maximum of two random variables i.i.d. drawn from $F(x)$ is $F_{\max}(x_m) = \Pr(x < x_m) \cdot \Pr(x < x_m) = [F(x_m)]^2$.

$$v(w_m) = \max_{a \in \{\text{Accept}, \text{Reject}\}} \left(\frac{w_m}{1 - \beta}, c + \beta \mathbb{E}_{\max w} [v(w'_m)] \right) \quad (1.2)$$

where $\mathbb{E}_{\max w} [v(w'_m)] = \int_0^B v(w'_m) d[F(w'_m)]^2$. Since $\mathbb{E}_{\max w} [v(w'_m)] = 2 \int_0^B v(w'_m) f(w'_m) F(w'_m) dw' > \int_0^B v(w'_m) f(w'_m) dw'$,¹ we have

$$\bar{w}_{\max} = (1 - \beta)(c + \beta \mathbb{E}_{\max w} [v(w'_m)]) > (1 - \beta)(c + \beta \mathbb{E} [v(w'_m)]) = \bar{w}.$$

■

1.3 Exercise 7.3

A McCall Model with n iid offers per period. Generalize the formulation in Exercise 7.2, we have

$$v(w) = \max_{a \in \{\text{Accept}, \text{Reject}\}} \left(\frac{w}{1 - \beta}, c + \beta \mathbb{E}_{\max w} [v(w')] \right) \quad (1.3)$$

¹We apply the monotonicity of $v(x)$ and $2F(x) - 1$ to finish the proof:

$$\begin{aligned} & 2 \int_0^B v(w'_m) f(w'_m) F(w'_m) dw' - \int_0^B v(w'_m) f(w'_m) dw' \\ &= \int_0^B v(w'_m) (2F(w'_m) - 1) dF(w') \\ &= \mathbb{E} [v(w'_m) (2F(w'_m) - 1)] \\ &= \underbrace{\text{Cov} [v(w'_m), 2F(w'_m) - 1]}_{>0} + \underbrace{\mathbb{E} [v(w'_m)] \mathbb{E} [2F(w'_m) - 1]}_{>0} \end{aligned}$$

where $\mathbb{E}_{\max w}[v(w')] = n \int_0^B v(w') F^{n-1}(w') f(w') dw'$.